

Practical 5

Title – Thermal Analysis – Transient Analysis (B)

Sample Example

Question: A fin of 6 mm diameter and 120 mm length is exposed to air at 30°C with convective heat transfer coefficient 10 W/m²K. The base temperature of the fin is 150°C. Find the temperature distribution with respect to time at various points on the fin and steady state heat transfer. Take the following properties for the fin material.

- Thermal conductivity (K) = 120 W/mK
- Density (ρ) = 8000 kg/m³
- Specific heat (C) = 380 J/kgK

Procedure

Step 1 Preferences

Open Ansys APDL – Preferences – Thermal – OK

Step 2 Pre-processor

Pre-processor – Element Type – Add – Solid – Tet 10 node 87 - OK - Close

Step 3 Material

Material Properties – Temperature Units – Celsius – OK

Material Properties – Material Models – Thermal – Conductivity – Isotropic

– Thermal Conductivity – Enter 120 W/cm °C – OK

Material Properties – Material Models – Thermal – Conductivity – Isotropic

– Specific Heat – Enter 380 j/kgk – OK

Material Properties – Material Models – Thermal – Conductivity – Isotropic

– Density – Enter 8000 – OK

Step 4 Modeling

Modeling – Create – Volumes – Cylinder – Solid Cylinder - Enter Radius 3 mm and length 120 mm – OK (select isometric view)

Step 5 Meshing

Meshing – Mesh Tool – Element Attribute – Volumes – Tick smart size – choose fine – (Check mesh – volumes – shape – tetrahedral) - Mesh – select object – OK

Click Raise Hidden – Close Mesh Tool Window

Step 6 Loads

Loads – Analysis Type – New Analysis – Transient – OK – Full - OK

Loads – Define Loads –Apply – Thermal – Temperature – On Areas – Select fixed side of cylinder – Ok – Choose Temperature – Enter Load Temperature Value as 150°C – OK

Loads – Define Loads –Apply – Thermal – Convection – on areas – select top, bottom and other face of cylinder– OK- Enter Vali Film Coefficient as 10 W/cm °C and enter Vali bulk temperature as 30°C – OK

Loads – Define Loads –Apply – Initial Condition – Define – Pick All – Select Temperature – Enter Temp 30 – OK

Step 7 Solution

Solution – Load Steps Opts – Output Control – DB/Result File – select every sub step – OK

Solution – Load Steps Opts – Time / Frequency – Time - time step – enter time at the load step end 8000 – OK

Stepped or ramped – selected stepped – minimum time step size – 0.01 – maximum time step size – 250 - OK

Solution – Solve –Current LS - OK

Step 8 General Results

General Post Process – Plot Results – Contour Plot – Nodal Solution – DOF – Nodal Temperature - OK

To See Animation

Plot Control – Animate – Deformed Results – Temperature – OK

General Post Process – Time Hist Postpro – Define Variable – Add – Normal DOF results – Ok – select any one-point cylinder – OK – OK (Repeat procedure 4-5 times) – Close

General Post Process – Time Hist Postpro – Graph Variables – Enter 2,3,4